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L02: Image Processing Fundamentals Reflective Journal

Techical Understanding:

This lab helped me better understand that a digital image is not really an “image” to a computer in the way that a person sees it. It is a numerical matrix made up of pixel values. I understood that concept in theory, but working through the operations made it much more concrete. Each change to the image, whether it involved brightness, contrast, cropping, blurring, or edge detection, was really just a mathematical operation being applied to those pixel values.

One of the techniques that interested me most was Gaussian blur. The name was familiar to me from previous math courses, and seeing it applied to image processing helped connect the mathematical concept to a practical use. The blur uses a Gaussian-weighted neighborhood around each pixel, giving more influence to nearby pixels and less to those farther away. What made this especially useful was seeing how Gaussian blur affected edge detection. When I applied Canny edge detection directly to my image, many small details in the brick, plants, and landscaping were detected as edges. After applying Gaussian blur first, many of those smaller details were reduced and the larger structural edges were easier to distinguish. Changing the Gaussian kernel from 15×15 to 5×5 also showed me how the amount of smoothing directly affects the result.

The most challenging techniques for me were multiplication and division. At first, multiplying or dividing one image by another seemed abstract and not very practical. It became clearer when I connected multiplication to masking. By multiplying an image by a binary mask, areas assigned a value of 1 remain visible while areas assigned 0 disappear. This could be useful when isolating a region of interest, such as a tumor in a medical image. Division also made more sense when viewed as a way to normalize uneven illumination rather than simply making an image darker. This could be useful in a scanned document or photograph where one area is much brighter than another.
Histogram analysis and contrast enhancement were also useful because my test image had strong sunlight in some areas and deep shadow in others. Global histogram equalization increased the overall contrast, but it made some of the bright areas appear harsh and seemed to accentuate the visual noise caused by sunlight. CLAHE produced a more natural-looking result because it enhanced contrast locally instead of applying one adjustment to the entire image. I could see more brick texture and structure in darker areas without making the brightest areas as aggressive. This was a good example of why a more adaptive method can be useful when an image has uneven lighting.

Connections and Applications:

The lab also helped me see how traditional image processing techniques still relate to modern AI image systems. Operations such as resizing, blurring, contrast adjustment, edge detection, and

normalization can all be used to prepare images before they are analyzed by machine-learning or deep-learning models. These techniques do not understand the meaning of the image, but they can improve the quality and consistency of the visual data being provided to an AI system. Modern tools such as Nano Banana go much further because they use learned models to interpret and modify image content, but the underlying information is still represented as pixels, channels, and numerical values.

Real-world applications for these techniques are easy to imagine. Gaussian blur can reduce image noise before edge detection. Thresholding can help separate objects from backgrounds. Image subtraction can help identify changes between two images, such as detecting a vehicle entering a previously empty driveway. Masking can isolate a specific region for analysis, and contrast enhancement can improve medical, security, or low-light images. These examples helped me understand that many basic image-processing operations are still useful even in more advanced computer vision systems.

Personal Reflection:

The aspect of image processing that interests me most is how combinations of relatively simple mathematical operations can produce meaningful changes in an image. Before this lab, I thought of image editing more from the user perspective: adjusting brightness, contrast, sharpness, or applying a filter. I now have a better understanding of what is happening numerically underneath those controls. The lab also made me more interested in how these traditional techniques are used as preprocessing steps in modern AI systems. One question I still have is how much preprocessing is typically performed manually in modern deep-learning computer vision systems compared with how much the model learns to compensate for on its own.

AI Disclosure: I used ChatGPT as a learning aid to help clarify unfamiliar image-processing concepts and organize my reflections based on the experiments I completed.